

PASHU SATHI — Technical & Architecture Q&A

Smart India Hackathon 2026 | Internal Evaluation Viva Guide (Technical Domain)

Problem Statement: SIH26128 — Early Detection & Prevention of Livestock Diseases	Core Stack: Spring Boot 3, Java 21, PostGIS 3.5, Flutter
Architecture: 3-Tier Micro-Modular with PostGIS Clustering & Redis 7	Validation: 68/68 Automated Tests Passing (100% Green)

1. System Architecture & High-Throughput Engineering

Q1: Walk us through your 3-Tier system topology and data flow.

Role: Backend / Cloud Engineer

- **Client Tier:** Flutter cross-platform mobile app (dual-mode Farmer/Vet) + React 18 TypeScript GIS Government Command Dashboard.
- **Application Tier:** Spring Boot 3 on Java 21 with Project Loom Virtual Threads for high-concurrency non-blocking I/O.
- **Data & Cache Tier:** PostgreSQL 17 + PostGIS 3.5 spatial extension for spatial indexing; Redis 7 for caching token sessions and weather metadata.
- **Cloud Infrastructure:** Hosted on an Azure Linux VM with Nginx reverse proxy, SSL termination, and Docker containerization.

Q2: How do you prevent database lock contention when thousands of field reports arrive at once?

Role: Backend / Database Engineer

- **Asynchronous Event Bus:** Disease intake endpoints return an immediate **201 Created** receipt; outbreak clustering logic executes asynchronously via Spring **@Async** threads.
- **Optimistic Locking (@Version):** Implemented on entity lifecycles to prevent stale state write-over without expensive pessimistic database row locks.
- **Spatial Query Offloading:** Heavy geospatial calculations are cached in Redis with a 30-minute TTL to keep PostGIS read load minimal.

2. AI Diagnostic Pipeline & Clinical Governance

Q3: How does your Gemini AI vision pipeline work, and how do you prevent hallucinations?

Role: AI / ML Engineer

- **Multimodal Visual Assessment:** Lesion photos are processed via Google Gemini Multimodal Vision API with constrained JSON schema output.
- **Strict Diagnostic Guardrails:** Prompts instruct the model to return differential probabilities, clinical confidence (0.00–1.00), and triage urgency.
- **Pre-Ingestion Quality Filter:** Image classifiers reject blurry images, non-livestock subjects, and corrupt payloads before invoking the model.
- **Strict Status Isolation:** AI output is permanently tagged **SUSPECTED** in the database; it is strictly advisory and cannot trigger legal quarantine on its own.

Q4: Explain the Human-in-the-Loop 'Vet Safety Gate'. Why is it legally necessary?

Role: AI Governance / Clinical Validation Dev

- **Statutory Compliance:** Under Indian Veterinary Council regulations, only a registered Veterinary Practitioner (VRC) can legally confirm an infectious disease.
- **Two-Phase State Machine:** Farmer/AI reports enter as **PENDING_REVIEW**. A licensed vet must verify physical clinical symptoms or lab assays.
- **False Alarm Blocker:** Outbreak detection algorithms only ingest verified veterinary reports (**VETERINARIAN** or **LAB_CONFIRMED**), completely eliminating false panics.

3. Geospatial Outbreak Engine & Multi-Signal Risk Scoring

Q5: How does your PostGIS spatial-temporal clustering algorithm detect outbreaks?

Role: Data / GIS Engineer

- **Spatial Distance Indexing:** Uses PostGIS `ST_DWithin(geography, geography, 25000)` backed by GiST R-Tree indexing for sub-millisecond proximity queries.
- **Temporal Sliding Window:** Evaluates contributing reports within a rolling 48-to-72-hour window (`createdAt >= NOW() - INTERVAL '48 HOURS'`).
- **Dynamic Centroid Calculation:** Outbreak centers are computed using `ST_Centroid(ST_Collect(location))` to update geographical cluster centers dynamically.
- **Zoonotic One-Case Trigger:** For lethal zoonoses (Rabies, Anthrax), the threshold drops to **1 confirmed case within 24 hours**, immediately creating a 50 km containment buffer.

Q6: Deconstruct the 4-Signal Composite Threat Score (0–100) used on your GIS map.

Role: Spatial Algorithms Engineer

- **Signal 1 — Sickness Severity (40%):** Ratio of confirmed vs. suspected cases, weighted by disease mortality and transmission R0.
- **Signal 2 — Live Weather Anomaly (20%):** Real-time temperature, relative humidity, and rainfall fetched from Open-Meteo API (favorable vector breeding ranges add score).
- **Signal 3 — Historical Baseline CV (20%):** Coefficient of variation comparing current incidence against 3-year seasonal historical baseline for that specific taluka.
- **Signal 4 — Vaccine Immunity Deficit (20%):** Estimated percentage of unimmunized susceptible livestock within the spatial perimeter.

4. Offline-First Sync & Anti-Fraud Security

Q7: How does the mobile app maintain full offline resilience in remote zero-network zones?

Role: Mobile / Flutter Dev

- **Local Persistent SQLite / Hive Storage:** Reports, audio recordings, and compressed image blobs are stored in an encrypted local database when offline.
- **Idempotent Background Worker:** Flutter background work-manager monitors network state. On network reconnection, queued reports sync automatically.
- **UUID-Based Deduplication:** Client generates deterministic v4 UUIDs for reports; backend enforces unique constraints so repeated retries never create duplicate tickets.

Q8: How do you prevent malicious users from spoofing disease reports to crash livestock prices?

Role: Full-Stack / Security Engineer

- **JWT Role-Based Access Control (RBAC):** Strict role isolation between **FARMER**, **VETERINARIAN**, and **GOVERNMENT_OFFICER** enforced via Spring Security 6 filter chains.
- **Tiered Confidence Scoring:** Unverified farmer reports carry a confidence weight of 0.20; they cannot mathematically trigger an outbreak cluster without vet confirmation.
- **Geo-Spoofing Mitigations:** Mobile client validates cell tower triangulation against device GPS; rapid teleportation triggers automated fraud review flags.